Work Sample data analysis 2018

In this Work Sample test you are expected to analyse and answer questions about the New York Citibike dataset, more information about the dataset is provided below. Briefly, the task you will complete is:

1. Explore and understand the dataset through queries
2. Provide insight about NYC by using the dataset creatively

We care about the solution and your approach, we don't care about the technology you use. We want you to use the right tool for the job. But do please let us know what you used as an FYI.

Some of the tasks are complex and most of them, except the algorithm test, are highly relevant for what we do at Unacast. For this test to be realistic, **we expect you to use Google / Stackoverflow etc. to guide you.** But please don't copy your solution of the web, then you're just fooling both of us :)

You can deliver your code and results as you feel like. We prefer if you use git and Github, and we'd appreciate that you don't publish your solution publicly. If you don't have a private repo on Github, Gitlab provides them for free for non-commercial use.

If you have any questions feel free to email andreas.heim@unacast.com. Please let us know if you for any reason need more time to complete this work sample test, and we'll work something out.

Good luck!

The dataset

You will be working with the public dataset from the New York Citibike. The dataset can be found on BigQuery, read more about it here, and requires you to setup a Google Cloud Platform account. Any justified expenses related to the case will be reimbursed by Unacast. However, Google will provide you with a 300 dollar free usage if you haven't used the platform before. The data is also available here if you want to set it up in your own database.

You will be using the two tables provided in BigQuery

1. Citibike_stations, and
2. Citibike_trips

station_id	INTEGER	REQUIRED	Unique identifier of a station.
name	STRING	NULLABLE	Public name of the station.
short_name	STRING	NULLABLE	Short name or other type of identifier, as used by the data publisher.
latitude	FLOAT	NULLABLE	The latitude of station. The field value must be a valid WGS 84 latitude in decimal degrees format.
longitude	FLOAT	NULLABLE	The longitude of station. The field value must be a valid WGS 84 longitude in decimal degrees format.
region_id	INTEGER	NULLABLE	ID of the region where station is located.
rental_methods	STRING	NULLABLE	Array of enumerables containing the payment methods accepted at this station.
capacity	INTEGER	NULLABLE	ANumber of total docking points installed at this station, both available and unavailable.
num_bikes_available	INTEGER	NULLABLE	Number of bikes available for rental.
num_bikes_disabled	INTEGER	NULLABLE	Number of disabled bikes at the station.
num_docks_available	INTEGER	NULLABLE	Number of docks accepting bike returns.
num_docks_disabled	INTEGER	NULLABLE	Number of empty but disabled dock points at the station.
is_installed	BOOLEAN	NULLABLE	Is the station currently on the street?
is_renting	BOOLEAN	NULLABLE	Is the station currently renting bikes?
is_returning	BOOLEAN	NULLABLE	Is the station accepting bike returns?
last_reported	TIMESTAMP	NULLABLE	Timestamp indicating the last time this station reported its status to the backend
eightd_has_available_keys	BOOLEAN	NULLABLE	Describe this field...
eightd_has_key_dispenser	BOOLEAN	NULLABLE	Describe this field...

Table 1: citibike_stations

tripduration	INTEGER	NULLABLE	Trip Duration (in seconds)
starttime	TIMESTAMP	NULLABLE	Start Time
stoptime	TIMESTAMP	NULLABLE	Stop Time
start_station_id	INTEGER	NULLABLE	Start Station ID
start_station_name	STRING	NULLABLE	Start Station Name
start_station_latitude	FLOAT	NULLABLE	Start Station Latitude
start_station_longitude	FLOAT	NULLABLE	Start Station Longitude
end_station_id	INTEGER	NULLABLE	End Station ID
end_station_name	STRING	NULLABLE	End Station Name
end_station_latitude	FLOAT	NULLABLE	End Station Latitude
end_station_longitude	FLOAT	NULLABLE	End Station Longitude
bikeid	INTEGER	NULLABLE	Bike ID
usertype	STRING	NULLABLE	User Type (Customer = 24-hour pass or 7-day pass user, Subscriber = Annual Member)
birth_year	INTEGER	NULLABLE	Year of Birth
gender	STRING	NULLABLE	Gender (unknown, male, female)

Table 2: citibike_trips

Task 1: Explore and understand the dataset

Use SQL to answer basic questions about the dataset below. In this task we expect you to provide us with a set of SQL queries and their results. We expect you to truncate data where it makes sense.

1. What is the unique number of stations in the dataset?
2. How many Citibikes is in the dataset?
3. What is the most popular start station and end station?
4. What is the gender distribution of the Citibike users?
5. What is the usertype distribution of the Citibike users?
6. What is the trip duration distribution of the Citibike trips?
7. What is the most popular Citibike trip?
8. Was there new bike introduced or removed at any point in time? What makes you think it did or didn't? What about the citibike stations?

Task 2: Popular areas in NYC

Can you estimate the most popular areas in NYC based on the Citibike dataset? Hint: You can use geohash to group latitude and longitudes.

It can be interesting to look at the dataset with different lenses, e.g.

- based on the time of day (e.g. during the morning, day, and evening)
- based on season
- based on gender
- combining multiple views

In this task we want you to focus on the presentation. Think of what is the best medium to explore the data, explain your findings, and answer the questions we provided. A tip is that you consider a mix of visualisations and raw numbers. The expected delivery for this task is either a report or a dashboard that you will present to us in the next interview.

NB: We expect that you use the dataset creatively. All assumptions doesn't have to make 100% sense but we expect you to be able to describe why you used the data in the way you did.